Beyond Action-Blind Prediction: Action-Sensitive Driving Video Generation with Diffusion Transformers

Maleeka Raddygala, Andrew Liang

Stanford Computer Science

Project Overview

The Problem: A pretrained video diffusion transformer [1] can generate plausible driving video from past frames alone, but that does not make it a controllable world model. For driving, we need the model to answer counterfactual questions: given the same observed scene, does the future change correctly if the ego-action sequence changes?

Our Objective Given 49 observed front-camera frames at 24 FPS, generate the next 72 frames while conditioning on future ego-action signals. The goal is not only visual fidelity, but measurable action dependence and correct-action alignment.

Main Finding: Compared to a no-action baseline, our action-conditioned model is no longer action-blind: changing the future ego-action sequence changes the generated future. At the same time, the model largely preserves the visual quality of the original video generator.

Core take away: Action control and visual fidelity are separate axes. Low-frequency action conditioning gives a practical path from action-blind video prediction toward action-sensitive driving world modeling.

Datasets & Metrics

- Pre-trained DiT: **LTX 2B 0.9.8 distilled [2]**
- 10 FPS front-camera driving clips derived from the **Waymo Open Dataset (WOD) dataset [3]**
- Each sample contains a upsampled **24 FPS video window** with **49 observed context frames** and **72 future frames to generate.**
- **7992 training, 50 validation, 1904 test windows**
- **Each frame consists of ego actions:**

Future ego trajectory	Past ego trajectory	Current state
Longitudinal position (20) Lateral position (20)	Velocity (x, y) (32) Acceleration (x, y) (32)	Speed (1) Angular Velocity (1) Camera Velocity (6)

$$x_{0:120} = (x_0, \ldots, x_{120}), \qquad a_{0:120} \in \mathbb{R}^{121\times 112}.$$

We evaluate two separate goals:

1. Visual quality: PSNR, SSIM, FVD [4, 5] distance, sharpness, motion ratio, and temporal consistency.

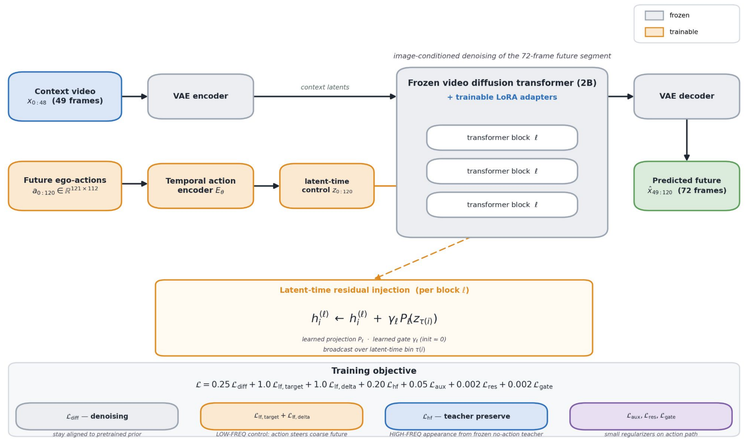
2. Action use: counterfactual action sensitivity and correct-vs-wrong action advantage.

$$S_{\mathrm{rgb}} = \frac{1}{3} \sum_{a' \in \mathcal{A}_{\mathrm{wrong}}} \mathrm{MAE}\big(G(a_{\mathrm{correct}}), G(a')\big)$$

$$\Delta\mathrm{PSNR} = \mathrm{PSNR}(G(a_c), x_{\mathrm{gt}}) - \frac{1}{3} \sum_{a' \in \mathcal{A}_{\mathrm{wrong}}} \mathrm{PSNR}(G(a'), x_{\mathrm{gt}}).$$

$$\Delta\mathrm{SSIM} = \mathrm{SSIM}(G(a_c), x_{\mathrm{gt}}) - \frac{1}{3} \sum_{a' \in \mathcal{A}_{\mathrm{wrong}}} \mathrm{SSIM}(G(a'), x_{\mathrm{gt}}).$$

Method



Training Objective

- Standard diffusion loss alone is insufficient action modules either had little effect or damaged the video prior.
- We therefore split the objective into low-frequency control and high-frequency preservation.

$$\mathcal{L} = 0.25\mathcal{L}_{\mathrm{diff}} + 1.00\mathcal{L}_{\mathrm{lf,target}} + 1.00\mathcal{L}_{\mathrm{lf,delta}} + 0.20\mathcal{L}_{\mathrm{hf}} + 0.05\mathcal{L}_{\mathrm{aux}} + 0.002\mathcal{L}_{\mathrm{res}} + 0.002\mathcal{L}_{\mathrm{gate}}.$$

$$\hat{u}_\theta = f_\theta(z_t, t, x_{\mathrm{ctx}}, a_{0:120}),$$

- Diffusion loss keeps the model on the pretrained video-generation objective.
- Low-frequency target loss makes the action branch predict the coarse future scene state.
- Low-frequency delta loss makes the model match how the scene changes over time.
- High-frequency teacher loss preserves edges, texture, and local appearance from the no-action visual model.
- Regularization keeps action residuals and gates small so actions do not overpower the visual prior.

$$\mathcal{L}_{\mathrm{diff}} = \frac{1}{|\mathcal{F}|} \sum_{i\in\mathcal{F}} \|\hat{u}_{\theta,i} - u_{t,i}\|_2^2, \qquad \bar{y}_\tau = \frac{1}{|\Omega_\tau|} \sum_{i\in\Omega_\tau} y_i,$$

$$\mathcal{L}_{\mathrm{lf,target}} = \frac{1}{|\mathcal{T}_\mathcal{F}|} \sum_{\tau\in\mathcal{T}_\mathcal{F}} \|\hat{\bar{u}}_{\theta,\tau} - \bar{u}_{t,\tau}\|_2^2, \qquad \Delta\bar{y}_\tau = \bar{y}_{\tau+1} - \bar{y}_\tau,$$

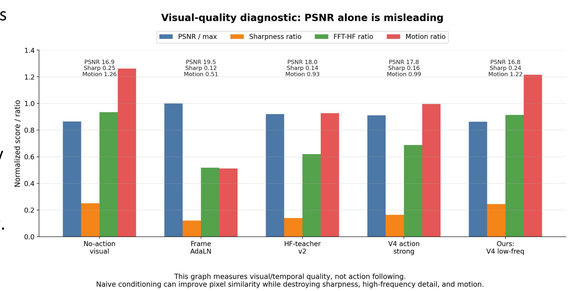
$$\mathcal{L}_{\mathrm{lf,delta}} = \frac{1}{|\mathcal{T}_\mathcal{F}| - 1} \sum_{\tau\in\mathcal{T}_\mathcal{F}} \|\Delta\hat{\bar{u}}_{\theta,\tau} - \Delta\bar{u}_{t,\tau}\|_2^2, \qquad \hat{u}^T = f_T(z_t, t, x_{\mathrm{ctx}}),$$

$$\mathrm{hf}(y_i) = y_i - \bar{y}_{\tau(i)}, \qquad \mathcal{L}_{\mathrm{hf}} = \frac{1}{|\mathcal{F}|} \sum_{i\in\mathcal{F}} \|\mathrm{hf}(\hat{u}_{\theta,i}) - \mathrm{hf}(\hat{u}_i^T)\|_2^2.$$

Experiments

- No-action preserves visual quality, but is not controllable.
- AdaLN has the best PSNR, but loses sharpness and motion.
- HF-teacher helps, but weakens detail and motion.
- V4 preserves sharpness, high-frequency detail, and motion closest to no-action.

Main takeaway: PSNR alone is misleading. Action conditioning must preserve the visual prior, not just improve pixel similarity.



Action Alignment Analysis

1. Counterfactual setup

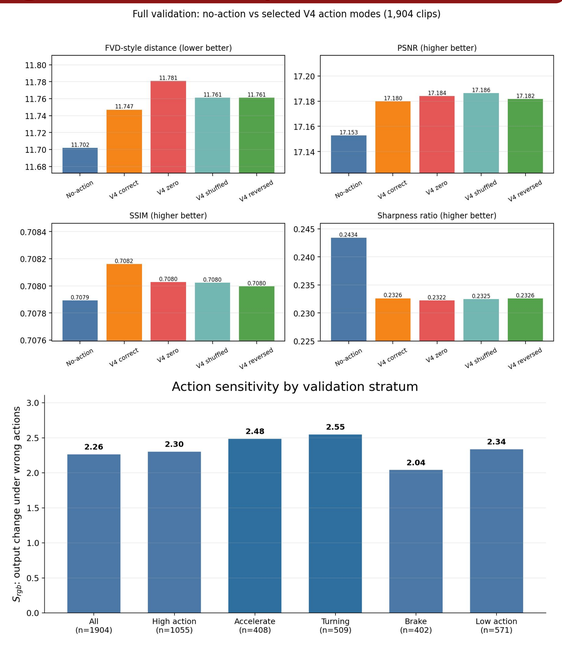
We keep the same 49 context frames and same random seed, then generate four futures:

- Correct: recorded future ego actions.
- Zero: normalized mean-action sequence.
- Shuffled: action sequence from another validation clip.
- Reversed: same context actions, but future actions reversed in time.

If the model ignores actions, all four outputs should be nearly identical.

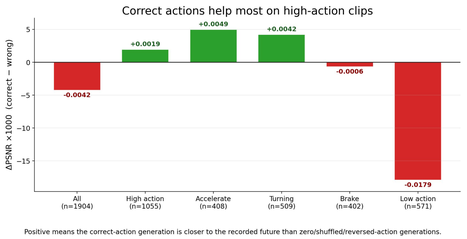
2. Action sensitivity

- Tests whether the model reacts when actions change.
- No-action baseline: zero sensitivity by construction.
- Our model: nonzero sensitivity across all strata.
- Strongest response: turning and acceleration.
- This proves action use, not correctness.



3. Correct-vs-wrong action advantage

- Tests whether correct actions beat wrong actions.
- Positive ΔPSNR means correct-action output is closer to ground truth.
- Correct actions help on high-action clips.
- Best results: acceleration and turning.
- Weak cases: braking and low-action clips.



Discussion & Future Research

Discussion:

- V4 improved action following (S_{rgb} from 0 to 2.262) without destroying PSNR, SSIM, FVD baselines
- ΔPSNR and ΔSSIM gains are small but positive. Further training could help
- Braking is especially difficult: Waymo deceleration is smooth and subtle, so pixel/structural metrics cannot distinguish incorrect braking from counterfactuals

Future Research:

- Try control fidelity metrics that can measure the success of action-conditioning using recovered motion instead of pixels
- Overweight/oversample/get more data for turning scenes

References

[1] W. Peebles and S. Xie. Scalable diffusion models with transformers. In Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023.
[2] Y. HaCohen, N. Chiprut, B. Brazowski, D. Shalem, D. Moshe, E. Richardson, E. Levin, G. Shiran, N. Zabari, O. Gordon, P. Panet, S. Weissbuch, V. Kulikov, Y. Bitterman, Z. Melumian, and O. Bibi. LTX-Video: Realtime video latent diffusion. arXiv preprint arXiv:2501.00103, 2024.
[3] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine, V. Vasudevan, W. Han, J. Ngiam, H. Zhao, A. Timofeev, S. Ettinger, M. Krivokon, A. Gao, A. Joshi, Y. Zhang, J. Shlens, Z. Chen, and D. Anguelov. Scalability in perception for autonomous driving: Waymo open dataset. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020.
[4] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli. Image quality assessment: From error visibility to structural similarity. IEEE Transactions on Image Processing, 13(4):600–612, 2004.
[5] Unterthiner, S. van Steenkiste, K. Kurach, R. Marinier, M. Michalski, and S. Gelly. Towards accurate generative models of video: A new metric and challenges. In International Conference on Learning Representations Workshop, 2019.